

A noncommutatively rigid Fano fivefold

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Abstract

We construct a Fano fivefold that has vanishing second (and first) Hochschild cohomology. Its category of coherent sheaves is therefore rigid: it has no infinitesimal deformations whatsoever.

We work over the field $\mathbb{C}$. Consider the multiprojective space $X = \mathbb{P}_a^2 \times \mathbb{P}_b^2 \times \mathbb{P}_c^2 \times \mathbb{P}_d^3$, and the complete intersection Z given by

$$(1) \quad \begin{aligned} a_0 b_0 + a_1 b_1 + a_2 b_2 &= 0 \\ b_0 d_0 + b_1 d_3 + b_2 (d_1 + d_2) &= 0 \\ c_0 d_0 + c_1 d_1 + c_2 d_2 &= 0 \\ c_0 d_1 + c_1 d_2 + c_2 d_3 &= 0. \end{aligned}$$

These equations define a section s of the vector bundle $\mathcal{E} := \mathcal{O}_X(1, 1, 0, 0) \oplus \mathcal{O}_X(0, 1, 0, 1) \oplus \mathcal{O}_X(0, 0, 1, 1)^{\oplus 2}$. The purpose of this note is to collect the following properties of Z .

Theorem A. *The complete intersection Z is a smooth projective Fano fivefold, with only diagonal non-zero Hodge numbers given by $(1, 4, 8, 8, 4, 1)$, for which*

$$(2) \quad H^0(Z, T_Z) = H^1(Z, T_Z) = H^0(Z, \bigwedge^2 T_Z) = 0.$$

Using the Hochschild–Kostant–Rosenberg decomposition for Hochschild cohomology and Kodaira vanishing, we see that (2) is equivalent to $HH^1(Z) = HH^2(Z) = 0$.

We will prove this using computer algebra, in particular we will use Macaulay2 [5] for computations with explicit coordinates, and PartialFlagVarieties.jl [3] for computations with zero loci of vector bundles. The code is available in [1]. It is classical that there are no examples of del Pezzo surfaces with vanishing second Hochschild cohomology, and for Fano 3-folds this follows from [2]. We have not found an example for Fano 4-folds.

Smooth Fano fivefold We first verify that the equations in (1) define a nonempty smooth complete intersection of codimension 4 in X . Let $S = \mathbb{Q}[a, b, c, d]$ be the polynomial ring with the $\mathbb{Z}^4$ -grading determined by the four factors of X , and let $I \subset S$ be the ideal generated by these equations. The following Macaulay2 code checks that Z has dimension 5 and is smooth.

```
needsPackage "MultiprojectiveVarieties";

S = QQ[a0,a1,a2, b0,b1,b2, c0,c1,c2, d0,d1,d2,d3,
  Degrees => {3:{1,0,0,0}, 3:{0,1,0,0},
    3:{0,0,1,0}, 4:{0,0,0,1}}];

I = ideal {
  a0*b0 + a1*b1 + a2*b2,
  b0*d0 + b1*d3 + b2*(d1+d2),
  c0*d0 + c1*d1 + c2*d2,
  c0*d1 + c1*d2 + c2*d3
};

Z = projectiveVariety I;
assert(dim Z == 5);
assert(dim singularLocus Z == -1);
```

By adjunction, the anticanonical bundle is

$$(3) \quad \omega_Z^\vee \cong (\omega_X^\vee \otimes (\det \mathcal{E})^\vee)|_Z \cong \mathcal{O}_Z(3-1, 3-2, 3-2, 4-3) = \mathcal{O}_Z(2, 1, 1, 1).$$

Since $\mathcal{O}_X(2, 1, 1, 1)$ is ample, its restriction to Z is ample, so Z is a Fano fivefold.

Cohomology of bundles The following Julia code computes the Hodge numbers in Theorem A, and computes the cohomologies of bundles we will use in what comes next.

```
using PartialFlagVarieties

X = product(projective_space(2), projective_space(2),
            projective_space(2), projective_space(3))
E = O(X, [1,1,0,0]) + O(X, [0,1,0,1]) + 2 * O(X, [0,0,1,1])
Z = zero_locus(E)

diagonal = [1,4,8,8,4,1]
@assert hodge_numbers(Z) == [p == q ? diagonal[p+1] : 0
                             for p in 0:5, q in 0:5]

T_XZ = restrict(Z, tangent_bundle(X))
N_ZX = normal_bundle(Z)
for (bundle, h0) in [(T_XZ, 39), (N_ZX, 39),
                    (exterior_power(T_XZ, 2), 582), (T_XZ * N_ZX, 1083)]
    H = cohomology(bundle)
    @assert H[0] == h0
    @assert all(H[q] == 0 for q in 1:5)
end

H = cohomology(tangent_bundle(Z))
@assert H[0] == H[1]
@assert all(H[q] == 0 for q in 2:5)
```

Setting up vanishing We now relate the bundles in the preceding cohomology computation to the tangent bundle of Z and its second exterior power. Let e_i denote the i th basis vector and write $(n_1, n_2, n_3, n_4) = (2, 2, 2, 3)$. If $p_i: X \rightarrow \mathbb{P}^{n_i}$ denotes the i th projection, then $T_X \cong \bigoplus_{i=1}^4 p_i^* T_{\mathbb{P}^{n_i}}$. Thus, taking the direct sum of the pullbacks of the Euler sequences and restricting to Z gives

$$(4) \quad 0 \rightarrow \mathcal{O}_Z^{\oplus 4} \xrightarrow{\varepsilon} \bigoplus_{i=1}^4 \mathcal{O}_Z(e_i)^{\oplus n_i+1} \rightarrow T_X|_Z \rightarrow 0,$$

whilst the normal bundle sequence is

$$(5) \quad 0 \rightarrow T_Z \rightarrow T_X|_Z \xrightarrow{ds} N_{Z/X} \rightarrow 0,$$

where $N_{Z/X} \cong \mathcal{E}|_Z$, and ds is induced by the Jacobian of s . The preceding cohomology computation in Julia gives

	$T_X _Z$	$N_{Z/X}$	$\wedge^2(T_X _Z)$	$(T_X _Z) \otimes N_{Z/X}$
(6) h^0	39	39	582	1083
$h^{\geq 1}$	0	0	0	0

The computation also gives $h^0(Z, T_Z) = h^1(Z, T_Z) = u \geq 0$, but we cannot yet determine the value of u . Since Z is Fano, Kodaira–Akizuki–Nakano vanishing gives $h^{\geq 2}(Z, T_Z) = 0$.

Vanishing for the tangent bundle The normal bundle sequence (5) then gives $u = 39 - \text{rank } H^0(Z, ds)$. The following Macaulay2 code continues the earlier snippet, and computes the rank of the Jacobian map from global sections of the middle term in (4) to $H^0(Z, N_{Z/X})$. The Koszul resolution of $\mathcal{O}_Z$ shows that the relevant global sections are computed by the degree-zero parts of the graded module presentations used in the code. Moreover,

the sections of the middle term in (4) surject onto $H^0(Z, T_X|_Z)$, since $H^1(Z, \mathcal{O}_Z) = 0$ by the Hodge-number computation above. Thus the rank computed by the code is the required rank of $H^0(Z, ds)$.

```
R = S/I;
F = R^(degrees S); -- middle term of the Euler sequence
N_ZX = R^(degrees source gens I); -- normal bundle
ds = map(N_ZX, F, substitute(transpose jacobian gens I, R));
A = substitute(matrix entries basis({0,0,0,0}, ds), QQ);
assert((numrows A, numcols A, rank A) == (39,43,39));
```

Thus $u = 0$, proving the cohomology vanishings for the tangent bundle.

Vanishing for the second exterior power of the tangent bundle The second exterior power of the normal bundle sequence (5) gives the exact sequence

$$(7) \quad 0 \rightarrow \bigwedge^2 T_Z \rightarrow \bigwedge^2 (T_X|_Z) \xrightarrow{\delta} (T_X|_Z) \otimes N_{Z/X} \rightarrow \text{Sym}^2 N_{Z/X} \rightarrow 0,$$

where, for local sections v, w of $T_X|_Z$, the map is given by $\delta(v \wedge w) = v \otimes ds(w) - w \otimes ds(v)$. Taking global sections identifies $H^0(Z, \bigwedge^2 T_Z)$ with $\ker H^0(Z, \delta)$, so it suffices to prove that $H^0(Z, \delta)$ is injective.

The following Macaulay2 code continues the earlier snippets, uses the presentation of $T_X|_Z$ provided by (4) to construct δ and its map on global sections, and checks that its matrix, of size 1083×582 by (6), has rank 582.

```
-- restricted Euler sequence
coordinates = flatten entries vars R;
epsilon = matrix apply(coordinates, x -> apply(degree x, weight -> weight*x));
T_XZ = coker map(F, R^4, epsilon); -- restricted tangent bundle

-- descend v tensor w |-> v tensor ds(w) - w tensor ds(v)
-- first to wedge^2 F, then to wedge^2(T_X|_Z).
onTensors = F ** ds - flip(N_ZX, F) * (ds ** F);
onWedges = quotient'(onTensors, wedgeProduct(1,1,F));
delta = inducedMap(T_XZ ** N_ZX, exteriorPower(2,T_XZ), onWedges);

-- map on global sections
B = substitute(matrix entries basis({0,0,0,0}, delta), QQ);
assert((numrows B, numcols B, rank B) == (1083,582,582));
```

Thus $H^0(Z, \delta)$ is injective, proving $H^0(Z, \bigwedge^2 T_Z) = 0$. This finishes the proof of Theorem A.

A blowup description We can also give a birational description of Z , by encoding the equations from (1) step-by-step using blowups.

Proposition 1 (Blowup description). *Let $C \subset \mathbb{P}_d^3$ be the twisted cubic defined by the maximal minors of (10), and set $V = \text{Bl}_C \mathbb{P}_d^3$ to be the Fano 3-fold with Mori–Mukai ID 2.27. Let $\Gamma \subset \mathbb{P}_a^2 \times V$ be the closure of the graph of the rational map on V induced by the projection*

$$(8) \quad \mathbb{P}_d^3 \dashrightarrow \mathbb{P}_a^2, \quad [d_0 : d_1 : d_2 : d_3] \mapsto [d_0 : d_3 : d_1 + d_2].$$

Then $\Gamma \cong \text{Bl}_p V$, where $p = [0 : 1 : -1 : 0] \notin C$. With this setup, there exists an isomorphism

$$(9) \quad Z \cong \text{Bl}_\Gamma(\mathbb{P}_a^2 \times V).$$

Proof. The twisted cubic C has a determinantal presentation: its homogeneous ideal is generated by the maximal minors of

$$(10) \quad \begin{pmatrix} d_0 & d_1 & d_2 \\ d_1 & d_2 & d_3 \end{pmatrix}.$$

By the standard determinantal description of the blowup of a twisted cubic [4, page 686], the last two equations in (1) cut out $V = \text{Bl}_C \mathbb{P}_d^3$. The map (8) is the projection from p . Since $p \notin C$, we identify p with its unique inverse image in V . The graph closure of the induced rational map on V is $\Gamma \cong \text{Bl}_p V$.

Write $(q_0, q_1, q_2) = (d_0, d_3, d_1 + d_2)$. On the chart $a_j \neq 0$, eliminate b_j from the first equation in (1). The second equation then becomes

$$(11) \quad \sum_{i \neq j} (q_i - a_i q_j / a_j) b_i = 0.$$

The two coefficients generate the ideal of Γ on this chart. Since Γ is smooth of codimension 2 in $\mathbb{P}_a^2 \times V$, they form a regular sequence, so (11) is the incidence equation for its blowup. These local identifications give $Z \cong \text{Bl}_\Gamma(\mathbb{P}_a^2 \times V)$. $\square$

Presumably, this description can also be used to prove Theorem A. For the Hodge numbers this follows from the blowup formula, whilst it is not clear to me how to show the anticanonical bundle is ample from this description. For the computations with the tangent bundle, it is probably feasible to prove that $H^0(Z, T_Z) = 0$, and thus $H^1(Z, T_Z) = 0$ by either the computation in `PartialFlagVarieties.jl` or a Grothendieck–Riemann–Roch computation. We have not tried to prove the vanishing of $H^0(Z, \wedge^2 T_Z)$ using this description.

On LLM usage LLMs were used to speed up the coding process of finding candidates and verifying the vanishing in an explicit example. The blowup description was suggested by the LLM.

References

- [1] Pieter Belmans. *A noncommutatively rigid Fano fivefold: computation code*. 2026. URL: <https://github.com/pbelmans/noncommutatively-rigid-fano-fivefold>.
- [2] Pieter Belmans, Enrico Fatighenti, and Fabio Tanturri. “Polyvector fields for Fano 3-folds”. In: *Math. Z.* 304.1 (2023), Paper No. 12, 30. DOI: 10.1007/s00209-023-03261-2. MR: 4578397.
- [3] Pieter Belmans and Javier Fernández Píriz. *PartialFlagVarieties.jl*. Julia package, version 0.4.0. 2026. URL: <https://github.com/HomogeneousTools/PartialFlagVarieties.jl>.
- [4] Lorenzo De Biase, Enrico Fatighenti, and Fabio Tanturri. “Fano 3-folds from homogeneous vector bundles over Grassmannians”. In: *Rev. Mat. Complut.* 35.3 (2022), pp. 649–710. DOI: 10.1007/s13163-021-00401-2. MR: 4482268.
- [5] Daniel R. Grayson and Michael E. Stillman. *Macaulay2, a software system for research in algebraic geometry*. URL: <https://macaulay2.com>.

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